

University of Dhaka
Affiliated Engineering Colleges
Department of Computer Science and Engineering
1st Year 1st Semester B.Sc. in CSE Final Examination, 2021
EEE-1103: Electrical Circuits

Total Marks: 70

Time: 3 Hours

(Answer any 5 (Five) of the following Questions)

1. a) On which factors the resistance of a material depends? Relate those factors to find the resistance of a material. 4
- b) State Kirchhoff's voltage law with an example. 2
- c) Determine the unknown resistances in Fig. 1.1 given the fact that $R_2 = 5R_1$ and $R_3 = (1/2)R_1$. 4

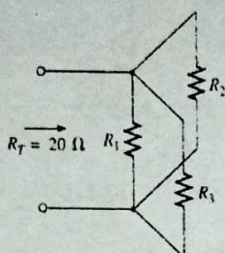


Fig. 1.1

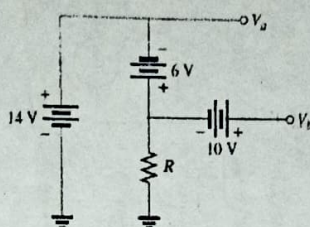


Fig. 1.2

$V = IR$
→

- d) For the network in Fig. 1.2, determine the voltages: V_a , V_b , V_{ab} . 4

2. a) State voltage divider rule with an example. 3
- b) "The total resistance of parallel resistors will always drop as new resistors are added in parallel, irrespective of their value"-explain. 3
- c) Given the information provided in Fig. 2.1, find the unknown quantities: E , R_1 , and I_3 . 4

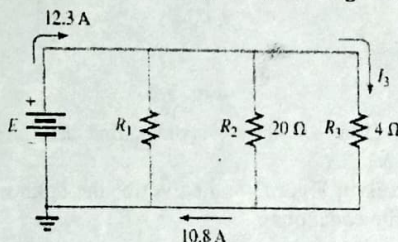


Fig. 2.1

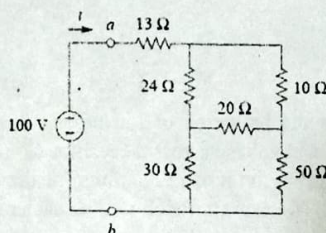


Fig. 2.2

- d) For the bridge network in Fig. 2.2, find R_{ab} using delta wye conversion and also find i . 4

3. a) State Kirchhoff's current law using a suitable figure. 2

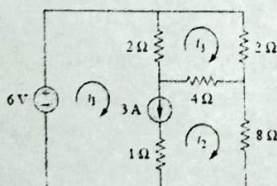


Fig. 3.1

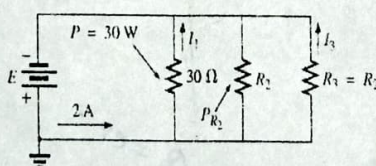


Fig. 3.2

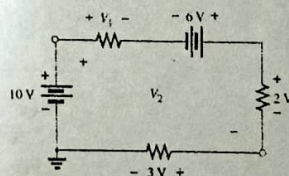


Fig. 3.3

- b) For the circuit in Fig. 3.1, use mesh analysis to determine I_1 , I_2 , I_3 . 4
- c) Find the unknown quantities for the network in Fig. 3.2 using the information provided. 5
- d) Using Kirchhoff's voltage law, find the unknown voltages for the configurations in Fig. 3.3. 3

$$V_x = \frac{R_x E}{R_T}$$

- 4 a) Find the voltage V_2 and the current I_1 for the network in Fig. 4.1.
b) Find Thevenin equivalent circuit of the following circuit in Fig. 4.2.

4
5

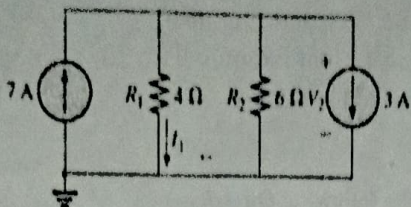


Fig. 4.1

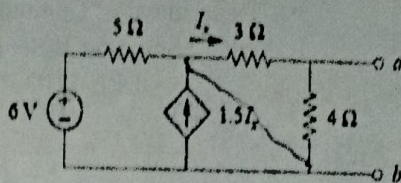


Fig. 4.2

- c) Using the supernode approach, determine the nodal voltages for the network of Fig. 4.3.

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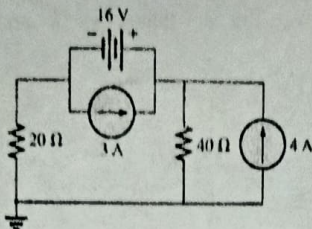


Fig. 4.3

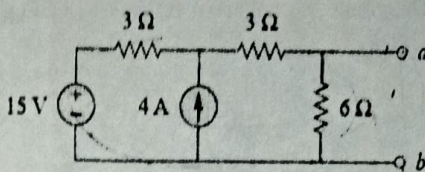


Fig. 5.1

5. a) Find Norton equivalent circuit of the above circuit in Fig. 5.1.

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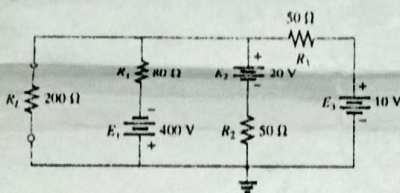


Fig. 5.2

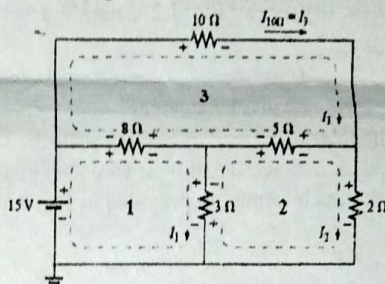


Fig. 5.3

- b) What is the limitation of Reciprocity theorem? Using Millman's theorem, find the current through and voltage across the resistor R_L in Fig. 5.2.
c) Write down the mesh equations for the network in Fig. 5.3 and solve for the unknown currents (format approach will be easier to write equations).

1+4

4

$$r = \frac{q}{V}$$

6. a) State Substitution theorem with an example.
b) Find the condition for maximum power and derive the equation for maximum power.
c) Use superposition theorem to find V_x in the following circuit in Fig. 6.1.

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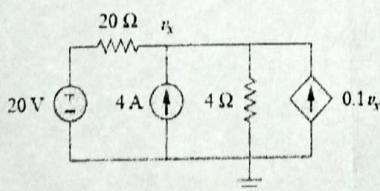


Fig. 6.1

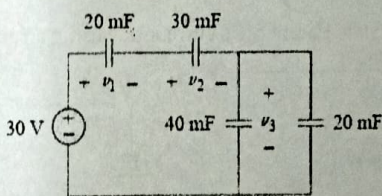


Fig. 7.1

$$\phi = eV$$

7. a) For Capacitor derive the equation $w = \frac{1}{2}Cv^2$.
b) Find the voltage across each of the capacitors in Fig. 7.1.
c) Define the following term:
i) Instantaneous Power ii) Average Power iii) Effective Value iv) Complex Power

5

5

4